

Exchange Argument

A simple way to prove a greedy choice is safe.

1. WHAT IS IT?

Take an optimal solution, replace its choice with the greedy choice, and show that the solution does not become worse.

2. WHY?

Greedy choices can look right, but we want to know whether they can destroy the optimal answer. This proof checks that.

3. DO I NEED IT TO CODE?

No. It is mainly a correctness proof.

Useful when the greedy choice is not obvious.

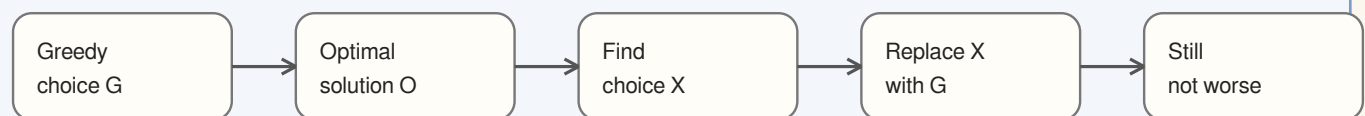
4. EXAMPLE: ACTIVITY SELECTION

Greedy rule: choose the activity that finishes earliest.



X -> G same number of activities + still valid = not worse

5. THE WHOLE IDEA



REMEMBER Greedy choice -> put it into an optimal solution -> nothing becomes worse -> greedy choice is safe.